Activity 3 Report: Cross-Group Deployment and Verification

COMP 018 Database Administration | Project tested: BoardTrack (Boarding House and Rental Management) (Node.js + PostgreSQL) | Date of testing: 6 November 2026

1. Cover and links

Item	Value
Testing group	G08: Borreta, Joshua; Bola, Martin Lawrence; Juatchon, Joaquin Phillip; Marcos, Karl Raphael; Pacheco Jr., Sancho; Vera Cruz, Euan Joseph
Project tested	BoardTrack (Boarding House and Rental Management)
Owning group	Group 11 (BoardTrack)
Repository	https://github.com/karlrobertc-art/BoardTrack
Tracker	https://github.com/EclipseChan/bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System/tree/main/other_group_tests
Evidence files	a3_deploy.txt, a3_app.txt, screenshots of the application

Only the repository and its documentation were used. Setup was done on the tester's own PostgreSQL server.

2. Test environment

Item	Value
Operating system	Windows 11
PostgreSQL	PostgreSQL 18.6 (x86_64 Windows), psql on PATH
Node.js	Node v26.13.0 and npm 11.20.0. The README asks for Node 20.12 or later
Packages	express, express-session, ejs, pg (94 packages installed, 0 vulnerabilities reported)
Database used	boardtrack_db (the name in .env.example)

3. Installation procedure and result

Step	Action	Result
1	Extracted the zip and read README.md and docs/	Clear guide with a quick test script. Two setup methods are offered
2	npm install	OK, 94 packages in 3 seconds
3	Copied .env.example to .env and filled DATABASE_URL and SESSION_SECRET	OK. The tester used the postgres account, as the README example suggests
4	npm start (automatic setup)	Printed: Created database "boardtrack_db", migrated 001_schema.sql, seeded 001_sample_data.sql, BoardTrack running at http://localhost:3000
5	npm run db:setup again	Printed: skip 001_schema.sql and "seed skipped (database already has data)".

		Nothing was duplicated, as the README promises
6	Other documented method: db/boardtrack_full_setup.sql run with psql on an empty database	OK, 8 tables, the same as the automatic method

4. Database verification

Check	Result
Tables	8: property, room, tenant, lease, payment, utility_bill, app_user and schema_migrations
Constraints	70 in total: 8 primary keys, 6 foreign keys, 3 unique, 15 check, 36 not null, 1 exclusion constraint (ex_lease_no_overlap, no two ACTIVE leases overlap on one room)
Triggers and view	trg_lease_sync_room on lease, trg_room_status_guard on room, view v_lease_summary
Extensions	btree_gist 1.8, pgcrypto 1.4
Indexes	16, including idx_lease_room, idx_lease_tenant, idx_payment_lease and idx_utility_lease
Sample data	app_user 3, property 2, room 6, tenant 4, lease 4, payment 6, utility_bill 4
Demo accounts	landlord (admin), caretaker (staff), juan (tenant). Passwords are stored with bcrypt through pgcrypto (prefix $2a$06$, cost 6)

5. Application roles checked

The README quick test script was run through HTTP calls and the main screens were checked in a browser (Figures 1 to 6).

Test	Result
Log in as landlord, caretaker and juan	Pass (Figures 1, 2, 5 and 6)
1. Add an ACTIVE lease that overlaps room 101	Pass. HTTP 400, "That room already has an ACTIVE lease overlapping these dates." (Figure 3)
2. Add a lease for room 103	Pass. Accepted, room 103 became OCCUPIED (set by the trigger)
3. Set that lease to TERMINATED	Pass. Room 103 returned to AVAILABLE
3b. Set room 101 to AVAILABLE by hand while it has an ACTIVE lease	Pass. Rejected: "This room has an ACTIVE lease, so it cannot be AVAILABLE."
4. Add a payment of 1,000	Pass. total_paid of lease 1 changed from 7,000.00 to 8,000.00
5. Delete a lease that has payments	Pass. Blocked by the foreign key (error 23503 in the app log), the page returns to the list
Reversed dates, negative rent, zero payment amount	Pass. All rejected by CHECK constraints with a friendly message
Invalid date text, duplicate tenant email	Pass. "Invalid date." and "Duplicate value..."
/rooms/abc/edit and /rooms/999999/edit	Pass. HTTP 404 "Record not found."
Not logged in, wrong password, SQL text in the username	Pass. Redirected to /login each time

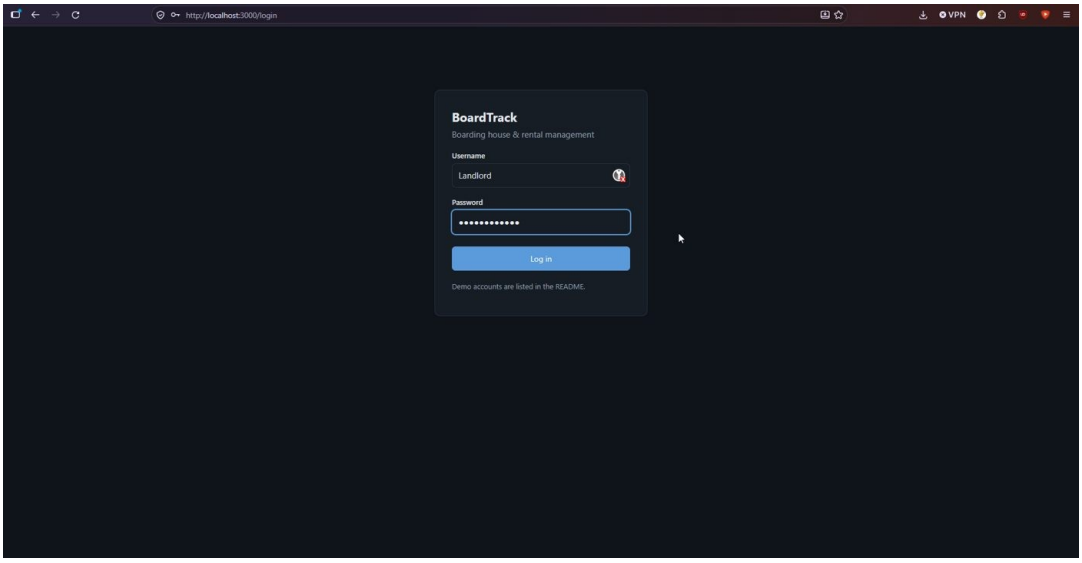


Figure 1. Login page.

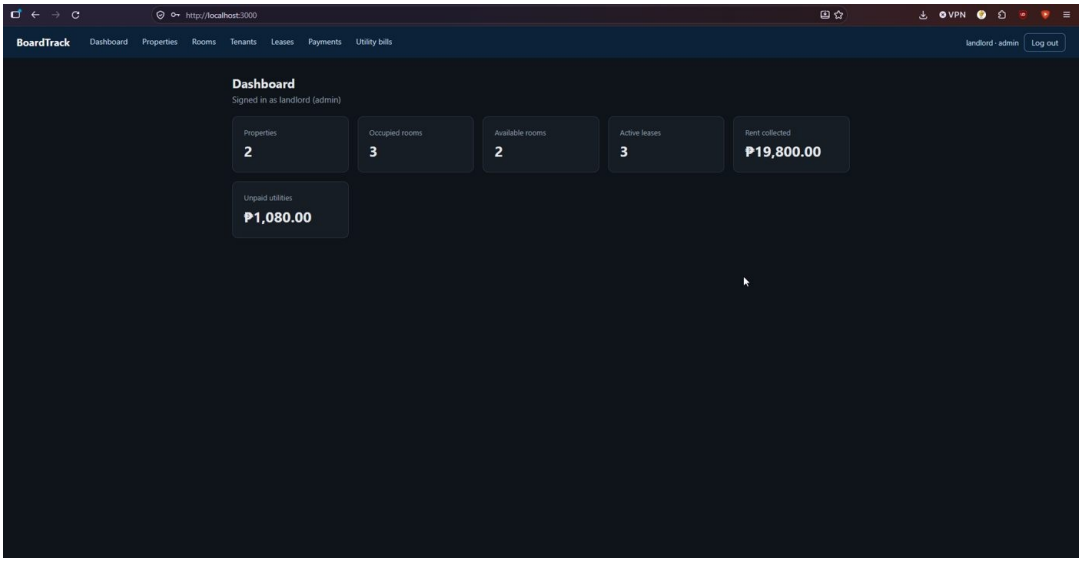


Figure 2. Landlord dashboard.

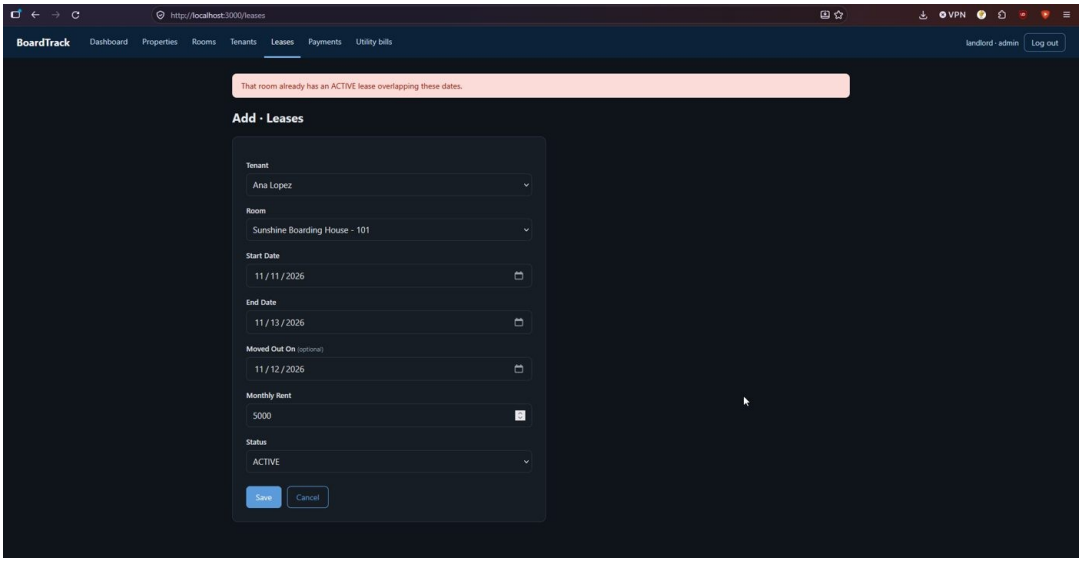


Figure 3. Overlapping lease rejected by the exclusion constraint.

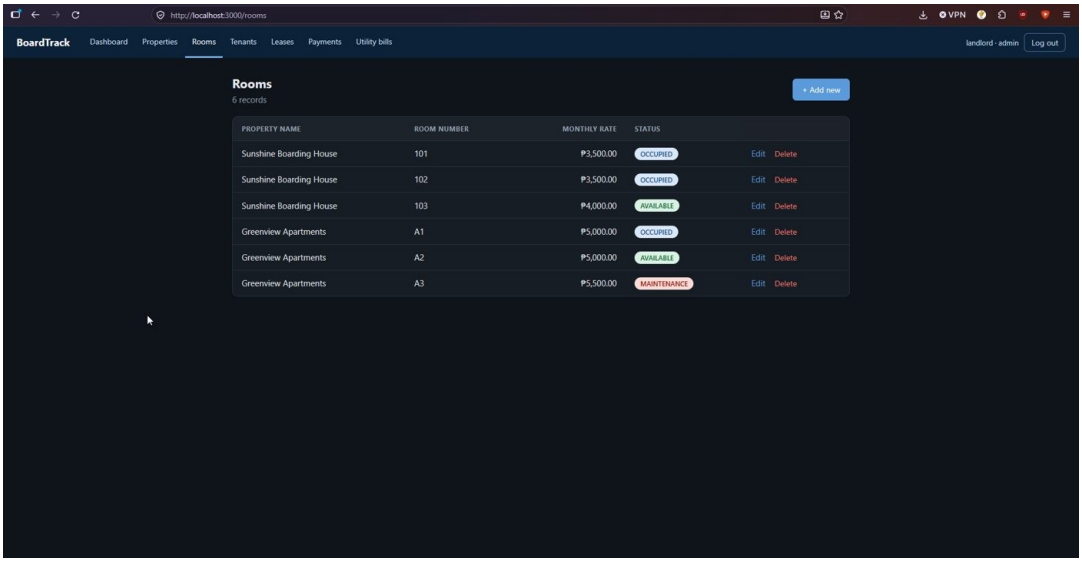


Figure 4. Rooms list with occupancy kept by the trigger.

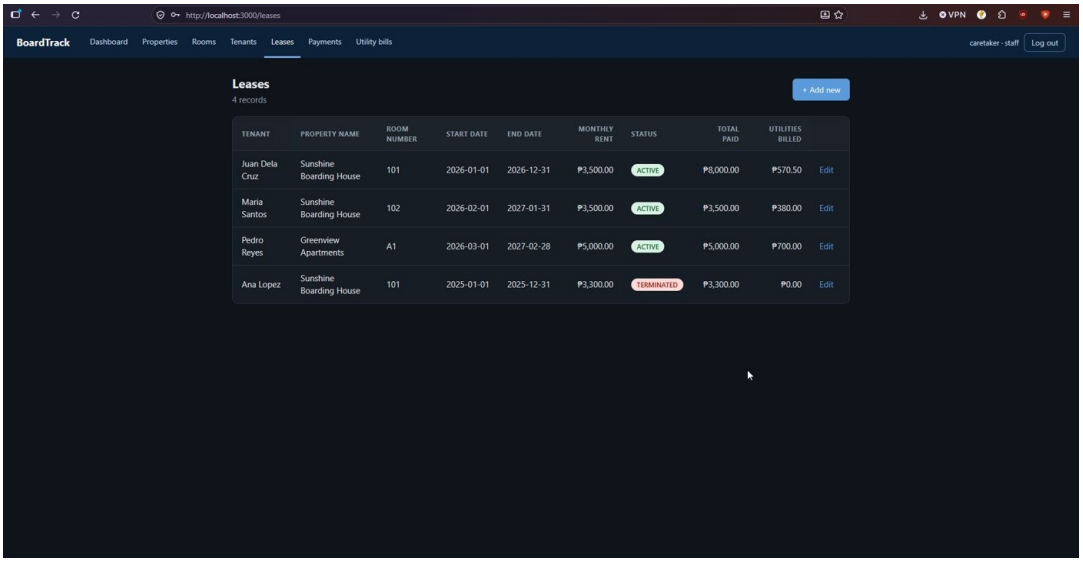


Figure 5. Caretaker view of Leases (no Delete buttons).

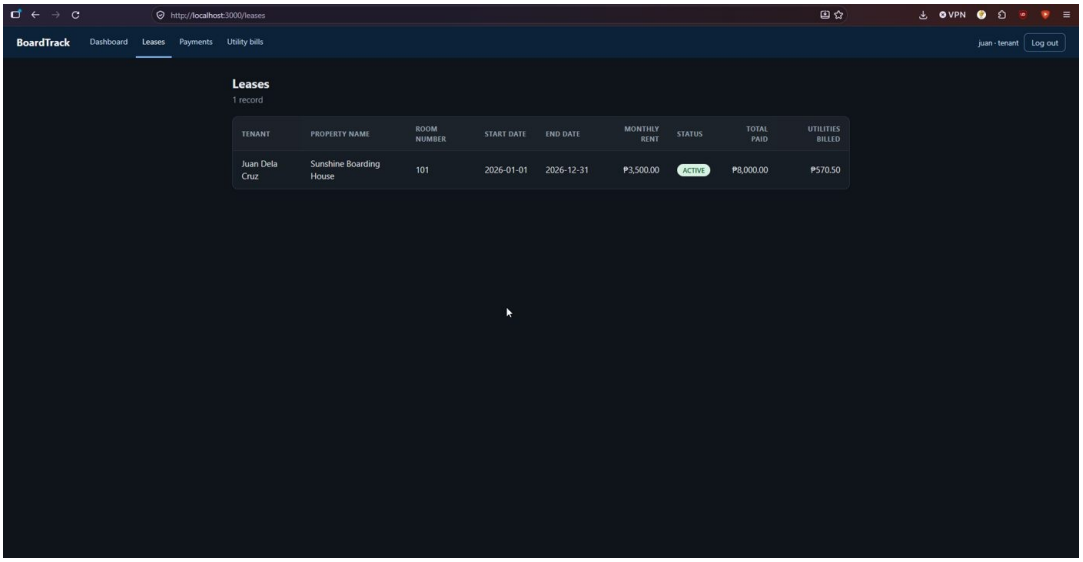


Figure 6. Tenant view of Leases (only his own lease).

6. Issues found

#	Issue	Impact and suggested fix
1	The app has no CSRF protection, no helmet headers and no login rate limiting. The response header X-Powered-By: Express is sent. The session cookie has HttpOnly but no Secure or SameSite attribute	A malicious page could trigger actions of a logged-in user, and passwords can be guessed without limit. Add csurf-style tokens, helmet and rate limiting, and set cookie flags
2	SESSION_SECRET falls back to a fixed text (dev-only-change-me) when the variable is missing, and sessions use the default in-memory store	Predictable secret if .env is forgotten; sessions are lost on restart. Fail at start-up when no secret is set
3	Demo passwords are the same for all accounts (ChangeMe123!) and are published in the README. The bcrypt	Acceptable for sample data. Use cost 10 or more for real accounts

	cost is 6 (the pgcrypto default)	
4	The README clone line uses the placeholder <REPOSITORY_URL> and the .env example uses placeholder credentials	Minor. Put the real URL in the README
5	Database access control is missing from the repository (see Activity 4)	The app runs as the account in DATABASE_URL, which the README example sets to postgres

What worked well: both setup methods worked first time, setup is repeatable, and every business rule is enforced in the database (exclusion constraint, two triggers, foreign keys, CHECK constraints) with clear messages.

7. Feedback sent to owners

Sent on 6 November 2026: issues 1 to 3 and 5. The setup was otherwise complete.

8. Reproducibility conclusion

Reproducible. A fresh install following the README gave a working system in one pass, both automatically and with the plain SQL file, and running setup again changed nothing.

AI usage declaration

- Tool: Claude (Anthropic).
- Tasks: interpreting the results of the testing.
- Validation: the commands were run by the group on its own server, and every result in this report comes from the saved output files.